



中国科学院空天信息创新研究院

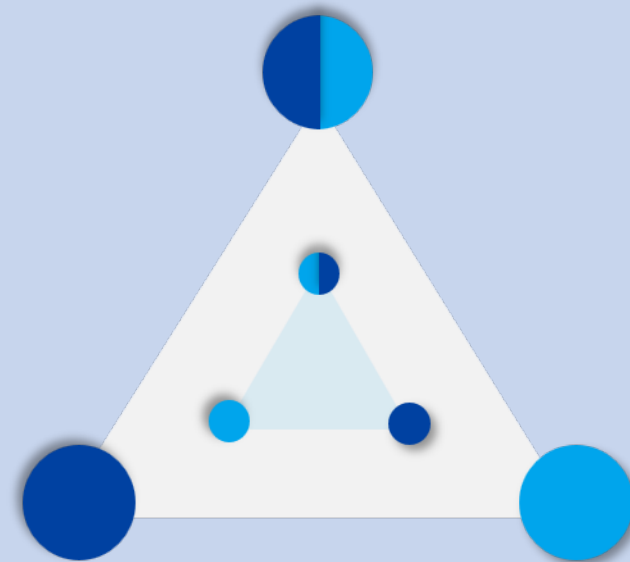
Aerospace Information Research Institute, Chinese Academy of Sciences

Photon Labeling

Photon Point Cloud Labeling & Visualization Software User Manual

Contact: Dr. Zhenwei Shi (shizw@aircas.ac.cn)

Download: <https://github.com/zwshi-pku/PhotonLabeling>

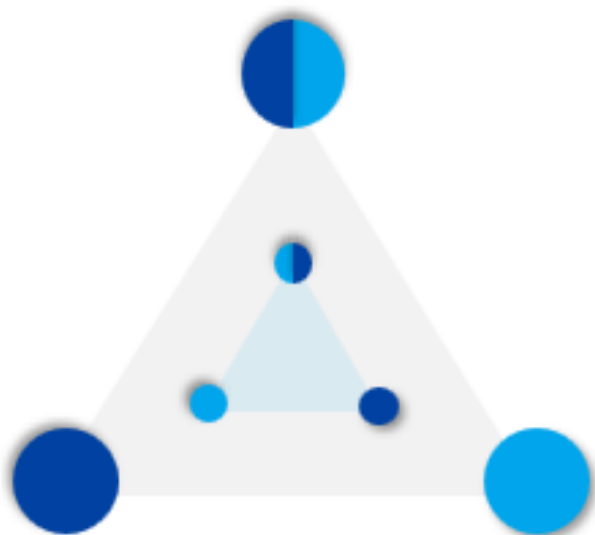




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The software is portable. After downloading, simply unzip the package and double-click the executable file ( PhotonLabeling.exe) to open the program and enter the start-up interface.



Photon Labeling

光子点云标注

Loading modules. / 正在加载模块.

Main Interface Overview

Menu Bar

Quick Settings Panel

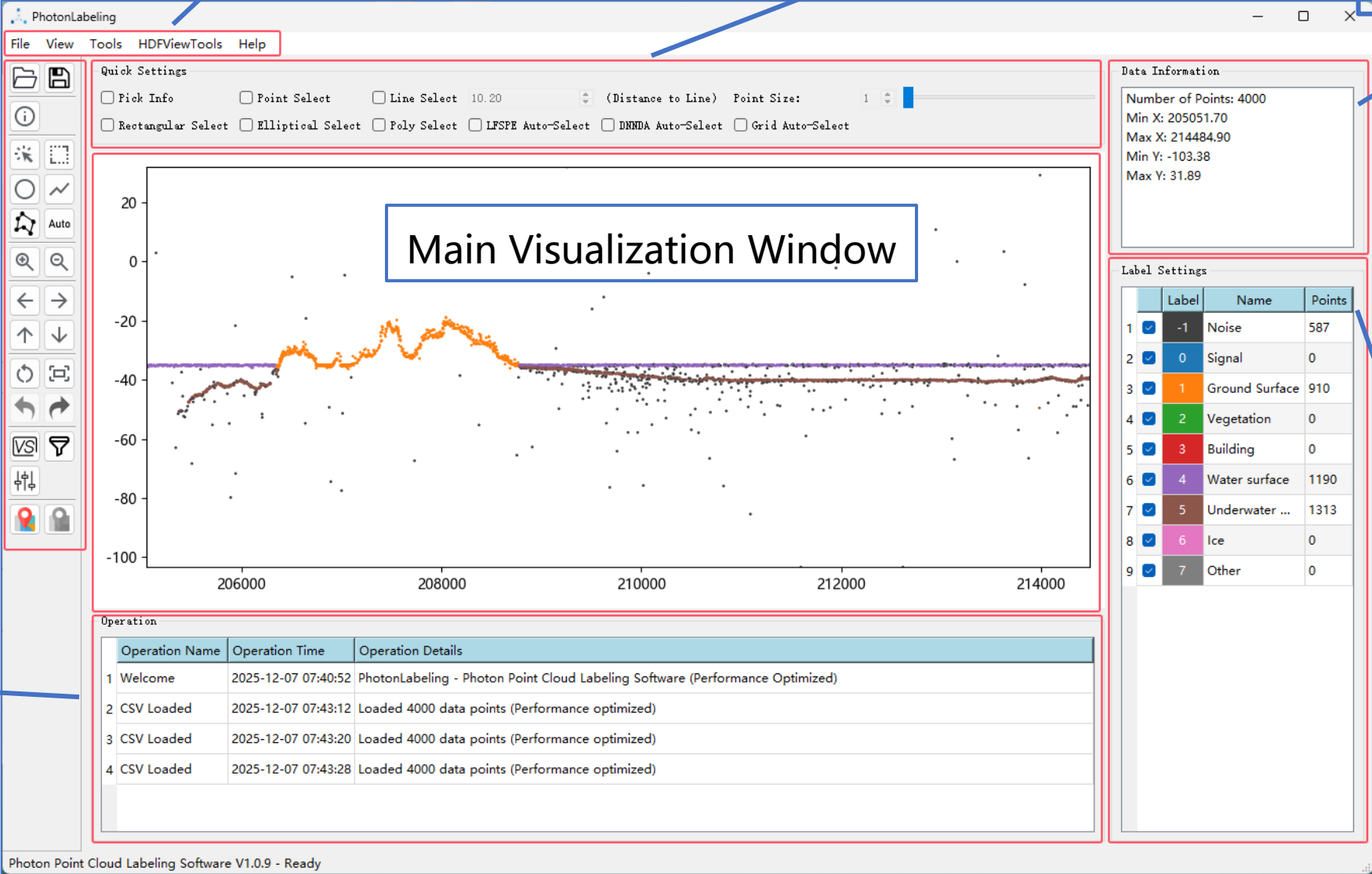
Point Set Overview Panel

Tool Bar

Main Visualization Window

Operation Log Panel

Label Settings Panel



The screenshot shows the PhotonLabeling software interface. At the top is a menu bar with 'File', 'View', 'Tools', 'HDFViewTools', and 'Help'. Below the menu bar is a tool bar with icons for file operations, selection, and visualization. To the right of the tool bar is a quick settings panel with checkboxes for 'Pick Info', 'Point Select', 'Line Select', 'Rectangular Select', 'Elliptical Select', 'Poly Select', 'LFSFE Auto-Select', 'DNMMA Auto-Select', and 'Grid Auto-Select'. It also includes a 'Distance to Line' slider and a 'Point Size' slider. The main visualization window displays a 2D scatter plot of point cloud data with a horizontal line and a wavy orange line. To the right of the main window is a point set overview panel showing 'Data Information' (Number of Points: 4000, Min X: 205051.70, Max X: 214484.90, Min Y: -103.38, Max Y: 31.89) and a 'Label Settings' table. At the bottom is an operation log panel showing a list of operations.

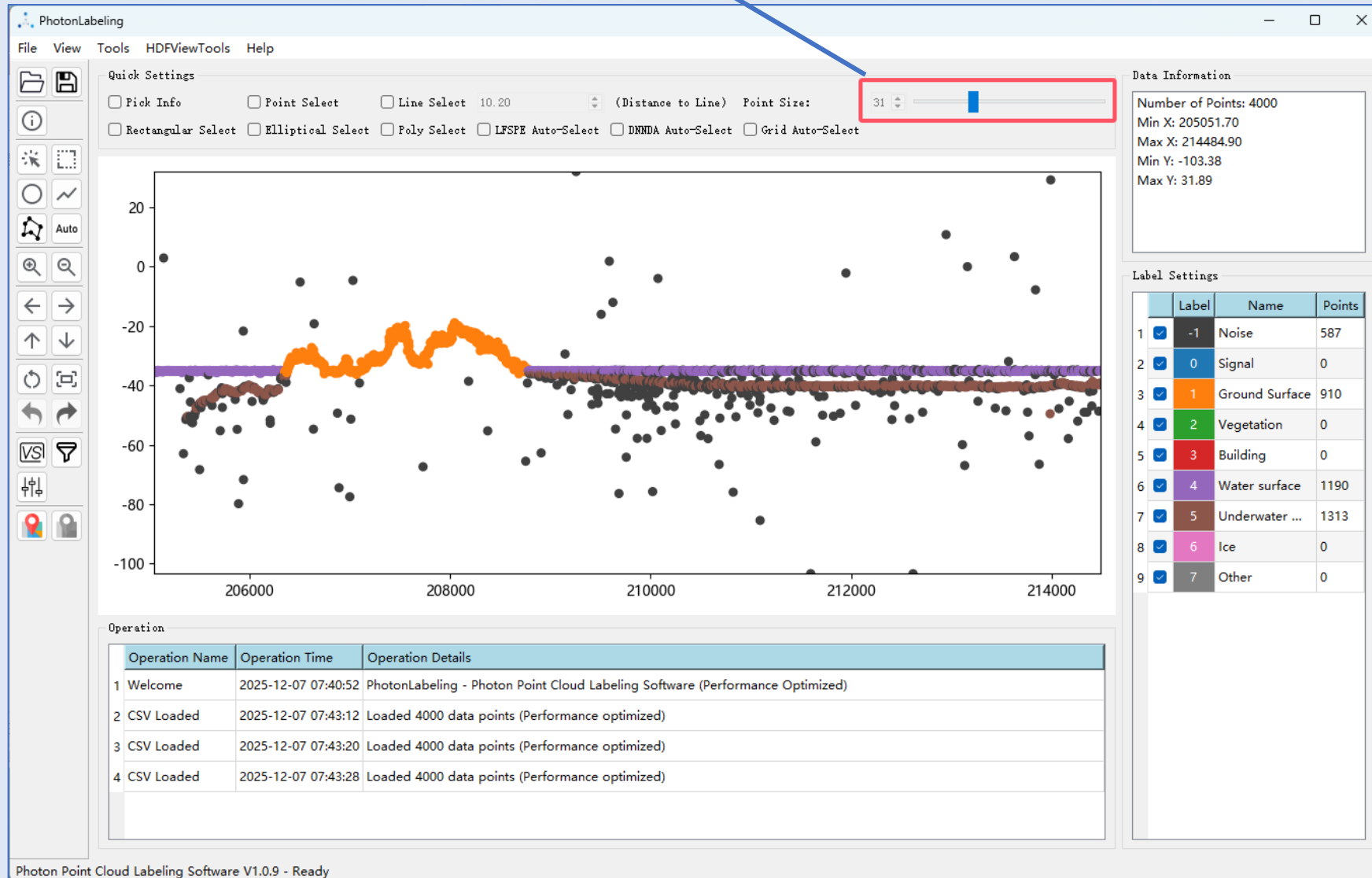
	Label	Name	Points
1	-1	Noise	587
2	0	Signal	0
3	1	Ground Surface	910
4	2	Vegetation	0
5	3	Building	0
6	4	Water surface	1190
7	5	Underwater ...	1313
8	6	Ice	0
9	7	Other	0

	Operation Name	Operation Time	Operation Details
1	Welcome	2025-12-07 07:40:52	PhotonLabeling - Photon Point Cloud Labeling Software (Performance Optimized)
2	CSV Loaded	2025-12-07 07:43:12	Loaded 4000 data points (Performance optimized)
3	CSV Loaded	2025-12-07 07:43:20	Loaded 4000 data points (Performance optimized)
4	CSV Loaded	2025-12-07 07:43:28	Loaded 4000 data points (Performance optimized)

Photon Point Cloud Labeling Software V1.0.9 - Ready

Tools and Functions

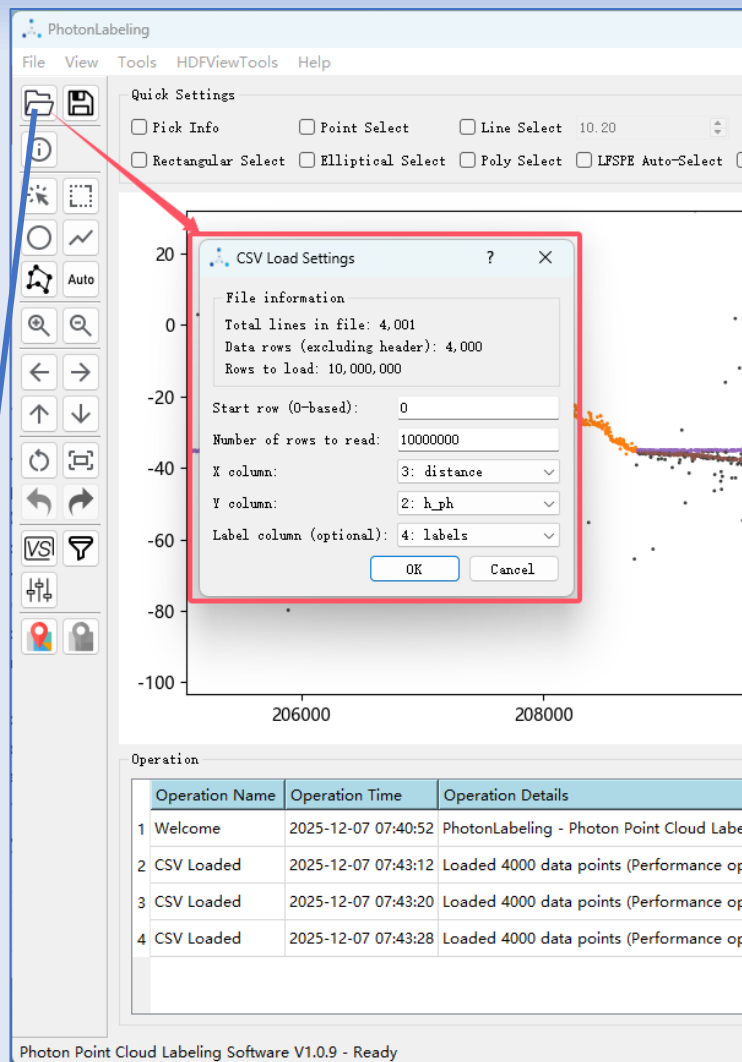
Point Size Control



CSV Import & Export

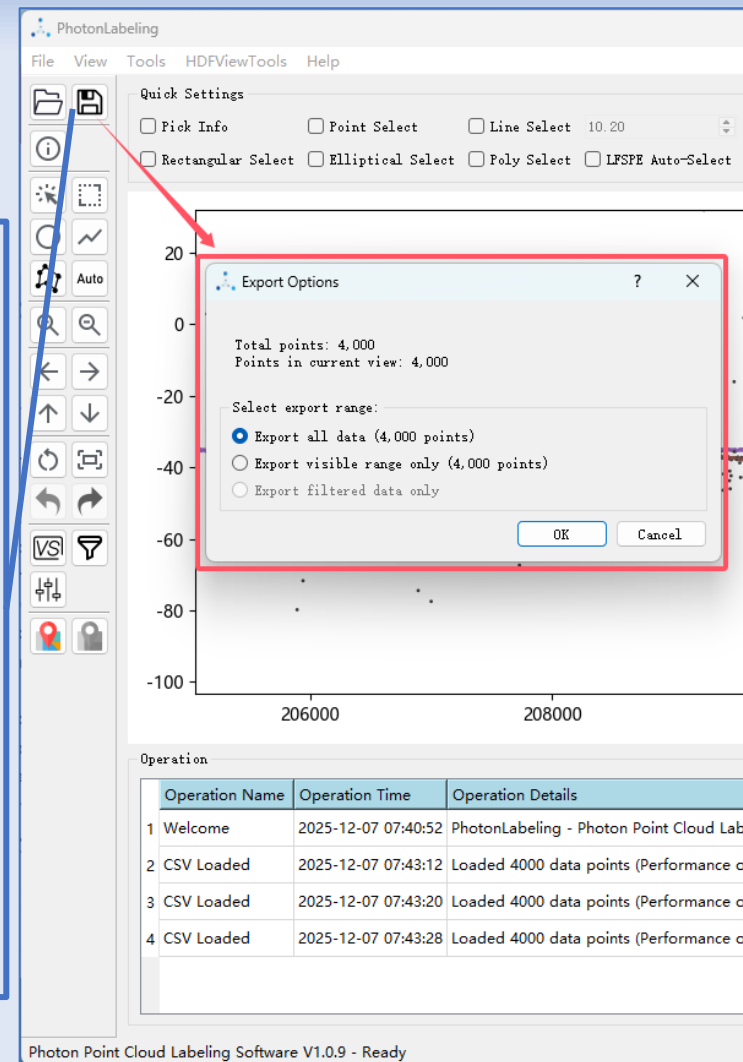
CSV Import

- ❑ Shortcut: Ctrl + O
- ❑ Supports importing more than 10 million points.
- ❑ If the file contains fewer than 10 million points, all points will be loaded.
- ❑ You can specify what each column represents (e.g., longitude, latitude, height, label, etc.).



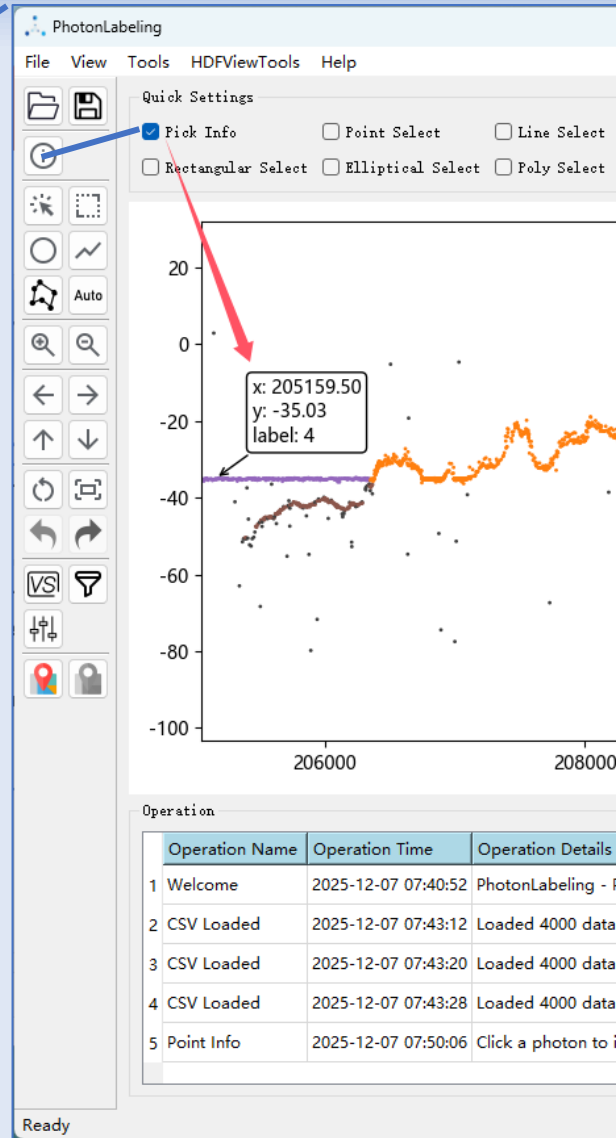
CSV Export

- ❑ Shortcut: Ctrl + S
- ❑ You can choose to export:
 1. All points
 2. Points within the current visible range
 3. Points filtered by label or other conditions
- ❑ If labels exist, the exported CSV will preserve the label values for each point.

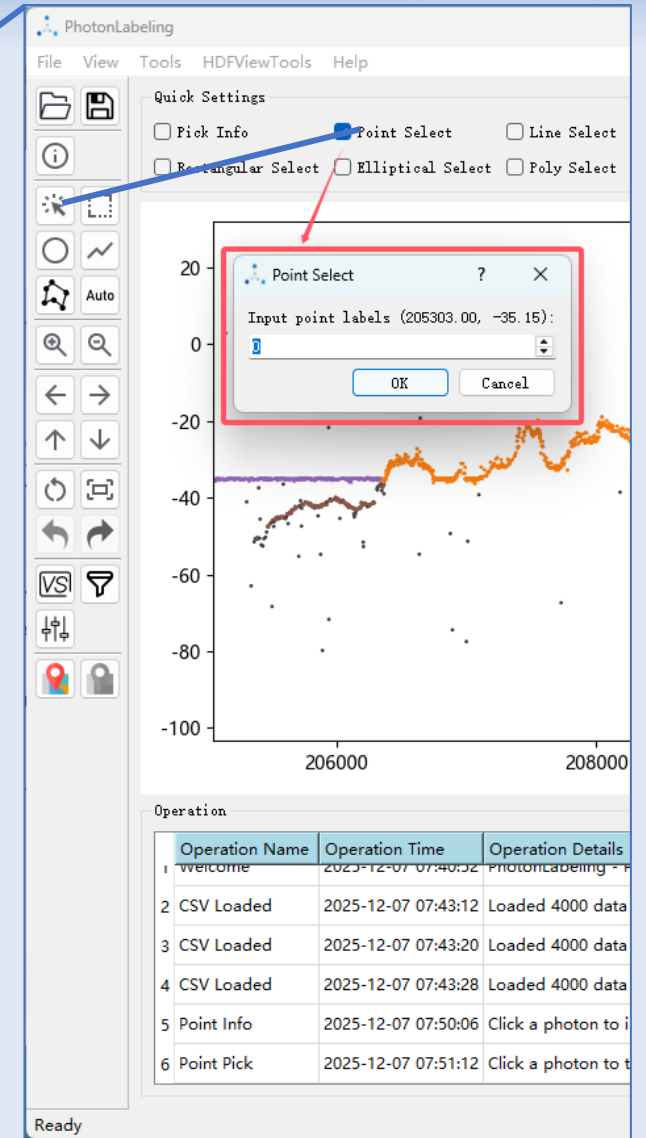


Tool: Point Info, Point Select

Check “Point Info”
When you click a photon point in the main window, the software displays the corresponding coordinates and label.



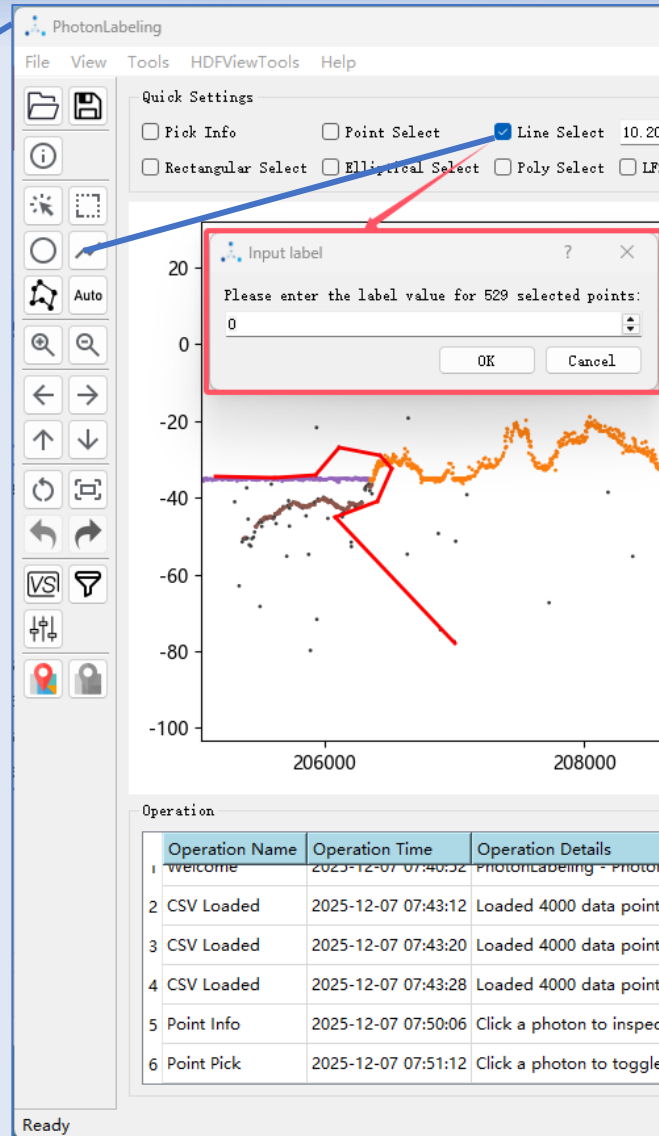
Check “Point Select”
When you click a photon point in the main window, a label setting dialog will pop up, allowing you to assign or modify the label for that point.



Tool: Line Select, Box Select

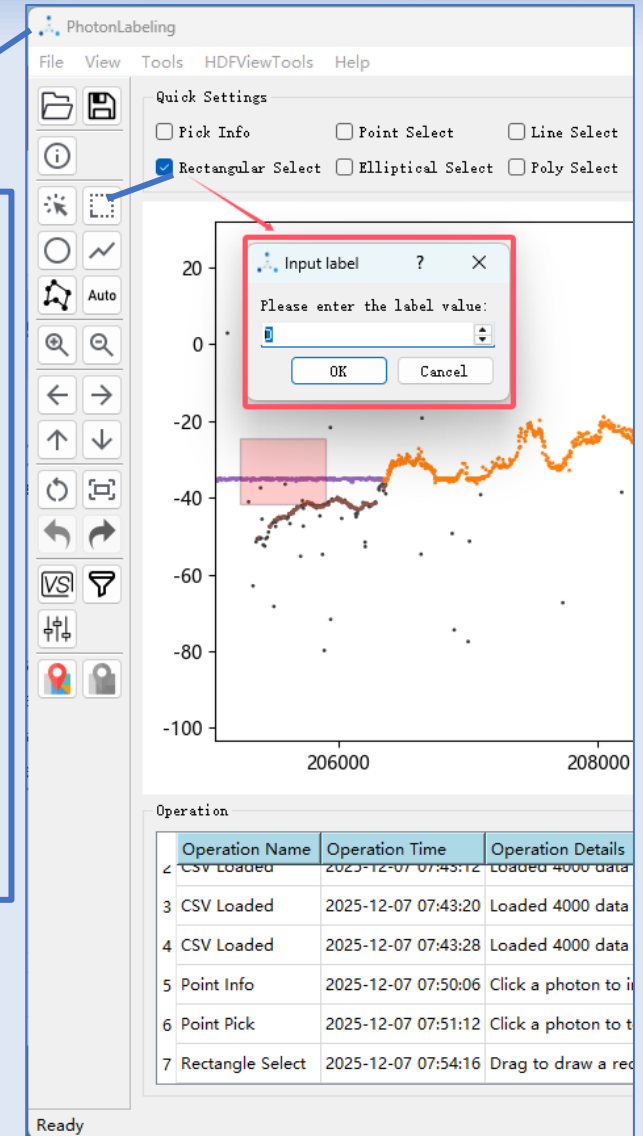
Check “Line Select”

Draw a polyline in the main window. The software will detect points near the polyline and assign them a specified label.



Check “Box Select”

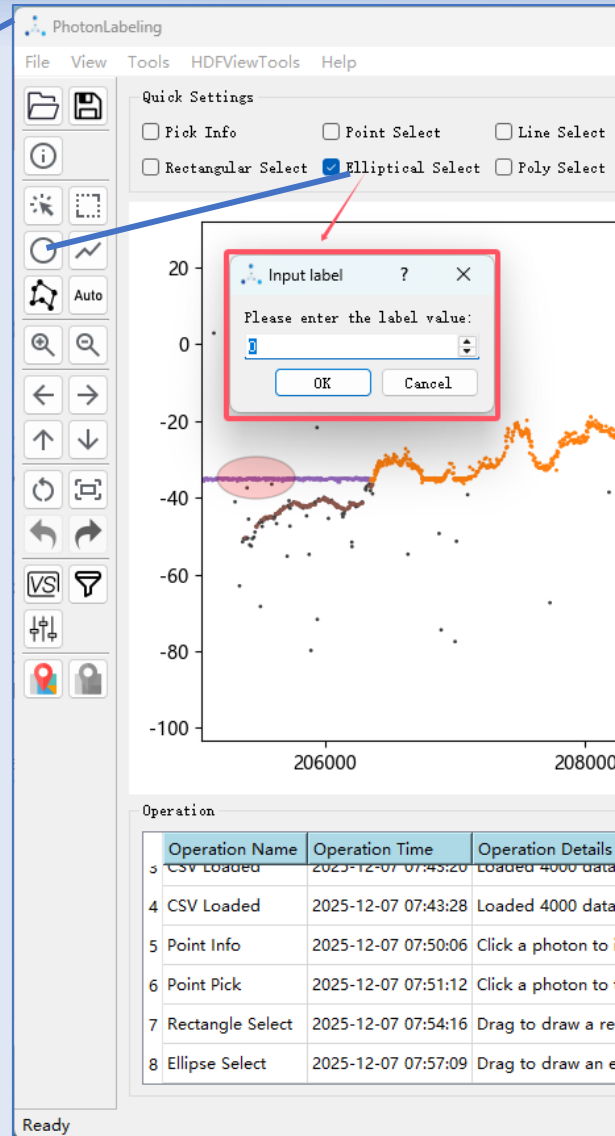
Drag to draw a rectangular region in the main window. All points within the rectangle will be assigned the same label.



Tool: Ellipse Select, Polygon Select

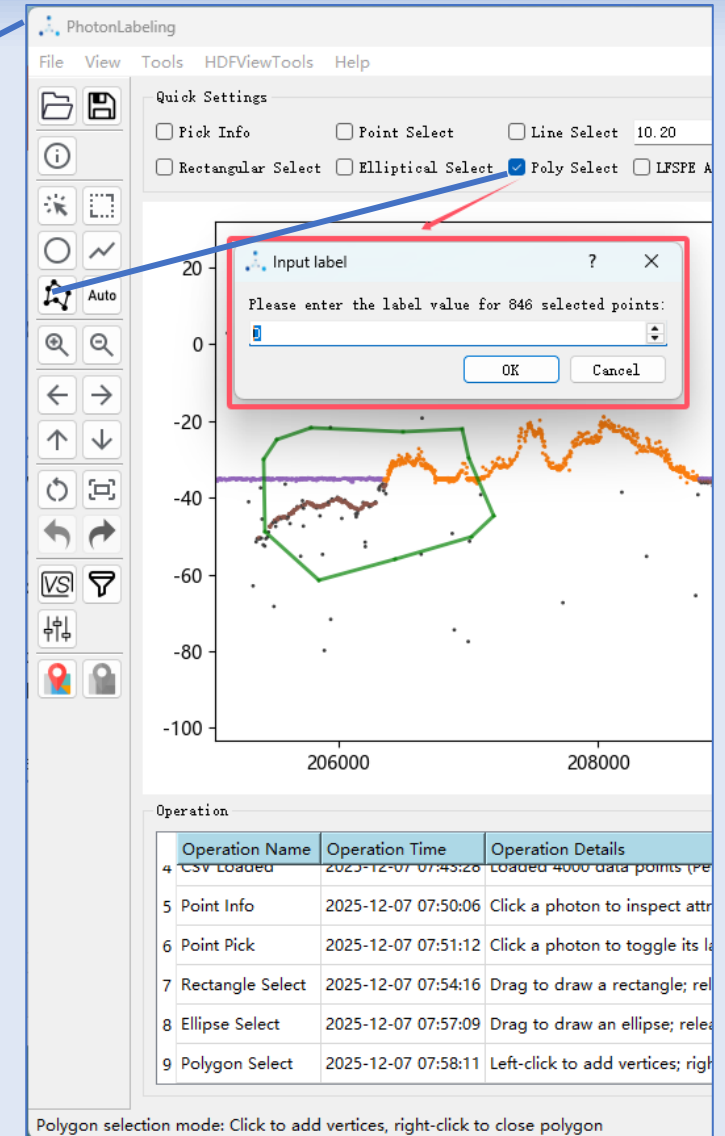
Check “Polygon Select”

Draw a polygon in the main window. Right-click to finish drawing. All points inside the polygon will be assigned the same label.



Check “Ellipse Select”

Drag to draw an elliptical region in the main window. All points inside the ellipse will be assigned the same label.



Tool: LFSPE Auto Selection

Check “LFSPE Algorithm Auto Selection” .

Drag a rectangle in the main window to select the photon subset that you want to auto-label.

The software will automatically invoke the LFSPE algorithm to perform labeling within this region.

The screenshot displays the PhotonLabeling software interface. The main window shows a 2D plot of photon data with a red rectangular selection box. The 'Quick Settings' panel at the top includes checkboxes for various selection methods, with 'LFSPE Auto-Select' checked. The 'Auto Labeling Parameters' dialog box is open, showing settings for automatic photon signal detection. The 'Basic Parameters' section includes 'Min Distance' (0.50 m), 'Search Radius' (30.0 m), 'Interval' (0.30), and 'Distance Threshold' (1.5 m). The 'Key Parameters' section includes 'Line Distance Threshold' (25.0 m) and 'Density Threshold' (10). The 'Label Settings' section includes 'Signal Photon Label' (1) and 'Noise Photon Label' (-1). The 'Threshold Selection Control' section has checkboxes for 'Use density threshold for signal detection' and 'Use distance threshold for signal detection', both of which are checked. The 'Visualization Options' section has a checkbox for 'Show density and distance comparison after labeling', which is unchecked. The 'Operation' table at the bottom shows a list of operations, including 'LFSPE Auto-Select'. The 'Data Information' panel on the right shows the number of points (4000) and the bounding box (Min X: 205051.70, Max X: 214484.90, Min Y: -103.38, Max Y: 31.89). The 'Label Settings' table on the right shows the results of the labeling process.

Label	Name	Points
-1	Noise	587
0	Signal	0
1	Ground Surface	910
2	Vegetation	0
3	Building	0
4	Water surface	1190
5	Underwater ...	1313
6	Ice	0
7	Other	0

Operation Name	Operation Time	Operation Details
Point Info	2025-12-07 07:30:00	Click a photon to...
Point Pick	2025-12-07 07:51:12	Click a photon to...
Rectangle Select	2025-12-07 07:54:16	Drag to draw a rectangle...
Ellipse Select	2025-12-07 07:57:09	Drag to draw an ellipse...
Polygon Select	2025-12-07 07:58:11	Left-click to add vertices...
LFSPE Auto-Select	2025-12-07 07:59:17	Drag a rectangle, set parameters, then run labeling on the selection.

In most cases, you only need to adjust: Line-to-Point Distance Threshold and Density Threshold to complete the labeling task. After the initial labeling is done, you can further fine-tune the results by adjusting these parameters via sliders, and the labeled results will update interactively. Note: For very large datasets, this algorithm may be relatively slow because it computes feature attributes for each photon.

Tool: DNND A Auto Selection

Check “DNND A Algorithm Auto Selection” .

Drag a rectangle in the main window to select the photon subset to be auto-labeled.

The software will automatically invoke the DNND A algorithm to label the selected photons.

The screenshot displays the PhotonLabeling software interface. The main window shows a scatter plot of points with a red rectangle highlighting a subset of points. A red arrow points from the 'DNND A Auto-Select' checkbox in the 'Quick Settings' panel to the 'DNND A - Settings' dialog box. The dialog box contains the following settings:

- Number of extraction: 1
- Number of nearest neighbors: 100 pts
- Number of distance groups: 20 groups
- Signal: 1

The 'Data Information' panel on the right shows the following statistics:

- Number of Points: 4000
- Min X: 205051.70
- Max X: 214484.90
- Min Y: -103.38
- Max Y: 31.89

The 'Label Settings' panel on the right shows a table of labels and their corresponding point counts:

	Label	Name	Points
1	☒ -1	Noise	587
2	☒ 0	Signal	0
3	☒ 1	Ground Surface	910
4	☒ 2	Vegetation	0
5	☒ 3	Building	0
6	☒ 4	Water surface	1190
7	☒ 5	Underwater ...	1313
8	☒ 6	Ice	0
9	☒ 7	Other	0

The 'Operation' panel at the bottom shows a list of operations performed:

Operation Name	Operation Time	Operation Details
0 Point Pick	2025-12-07 07:31:12	Click a photon to toggle its label.
7 Rectangle Select	2025-12-07 07:54:16	Drag to draw a rectangle; release to apply. Right-click to cancel.
8 Ellipse Select	2025-12-07 07:57:09	Drag to draw an ellipse; release to apply. Right-click to cancel.
9 Polygon Select	2025-12-07 07:58:11	Left-click to add vertices; right-click to close polygon.
10 LFSPE Auto-Select	2025-12-07 07:59:17	Drag a rectangle, set parameters, then run labeling on the selection.
11 DNND A Auto-Select	2025-12-07 08:00:21	Drag a rectangle to run labeling immediately on the selection; right-click to cancel.

The status bar at the bottom indicates 'Ready'.

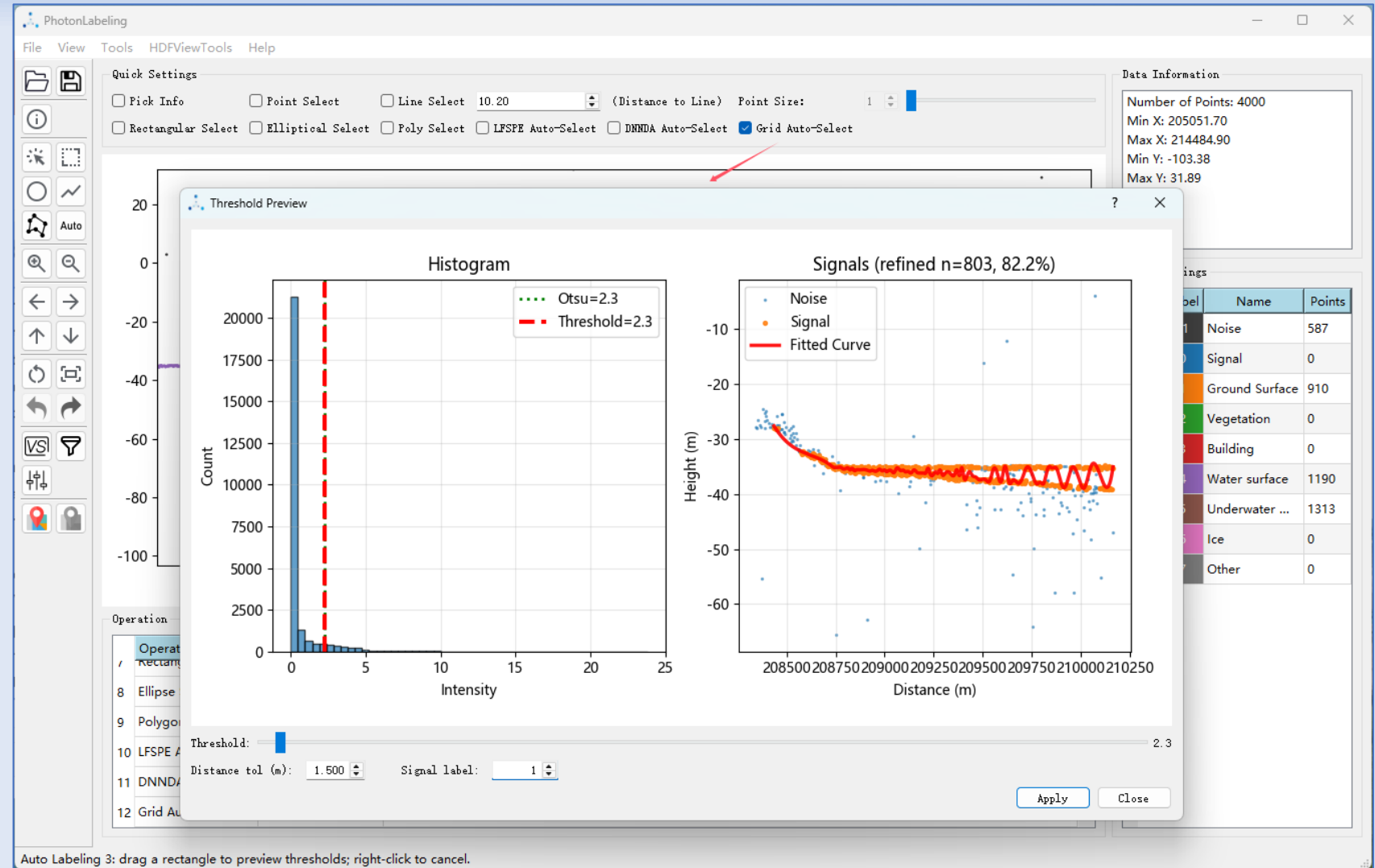
Tool: Grid Auto Selection

Check “Grid Algorithm Auto Selection” .

Drag a rectangle in the main window to select the photon subset to be auto-labeled.

The software will automatically invoke the Grid algorithm to perform labeling.

You can use the “Density Threshold” slider to quickly inspect and adjust the performance of signal photon detection.



Tool: Label Editing

You can merge labels or create new ones by editing label values.

Example: Change label 4 → 5

All points with label 4 will be merged into label 5.

Example: Change label 4 → 8

A new label 8 will be created, and all points previously labeled 4 will be reassigned to label 8.

Label Settings				
		Label	Name	Points
1	☒	-1	Noise	587
2	☒	0	Signal	0
3	☒	1	Ground Surface	910
4	☒	2	Vegetation	0
5	☒	3	Building	0
6	☒	4	Water surface	1190
7	☒	5	Underwater ...	1313
8	☒	6	Ice	0
9	☒	7	Other	0

Right-click a label that already has labeled points to open a context menu with operations relevant to existing labels.

Label Settings				
		Label	Name	Points
1	☒	-1	Noise	587
2	☒	0	Signal	0
3	☒	1	Ground Surface	910
4	☒	2	Vegetation	0
5	☒	3	Building	0
6	☒	4	Water surface	1190
7	☒	5	Underwater ...	1313
8	☒	6	Ice	0
9	☒	7	Other	0

Add Label

Change Label Value

Delete Label (has data, cannot)

Reset All Label Names

Right-click a label that has no labeled points to open a context menu tailored to empty labels (e.g., rename, delete, etc.).

Label Settings				
		Label	Name	Points
1	☒	-1	Noise	587
2	☒	0	Signal	0
3	☒	1	Ground Surface	910
4	☒	2	Vegetation	0
5	☒	3	Building	0
6	☒	4	Water surface	1190
7	☒	5	Underwater ...	1313
8	☒	6	Ice	0
9	☒	7	Other	0

Add Label

Change Label Value

Delete Label

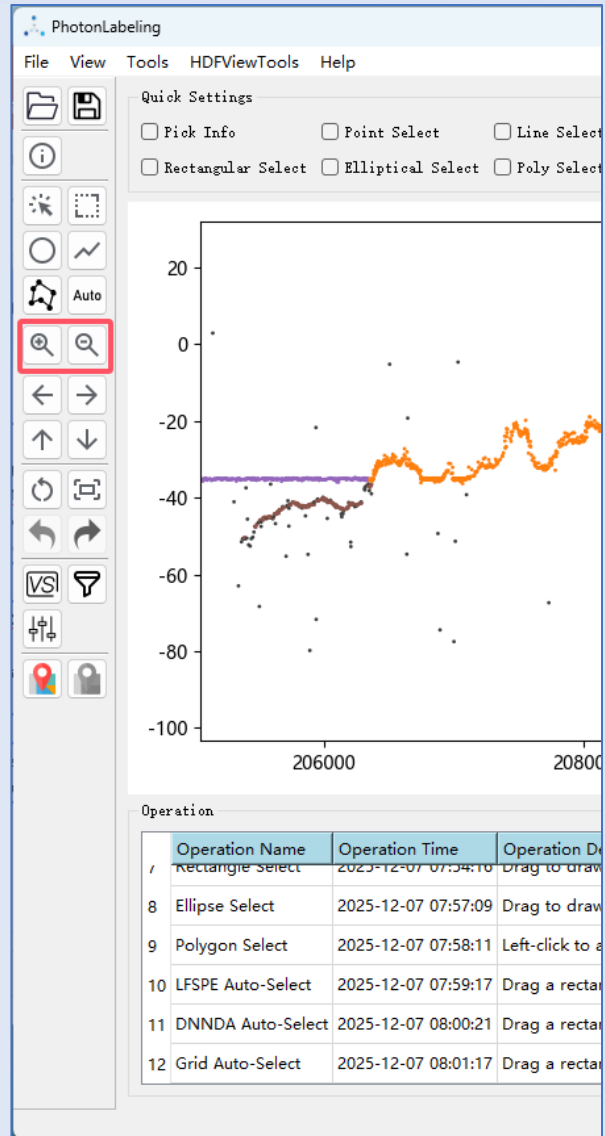
Reset All Label Names

Tool: Zoom In, Zoom Out, Move

Click “Zoom In” , then drag a rectangle in the main window. The selected region will be enlarged to fill the main view.

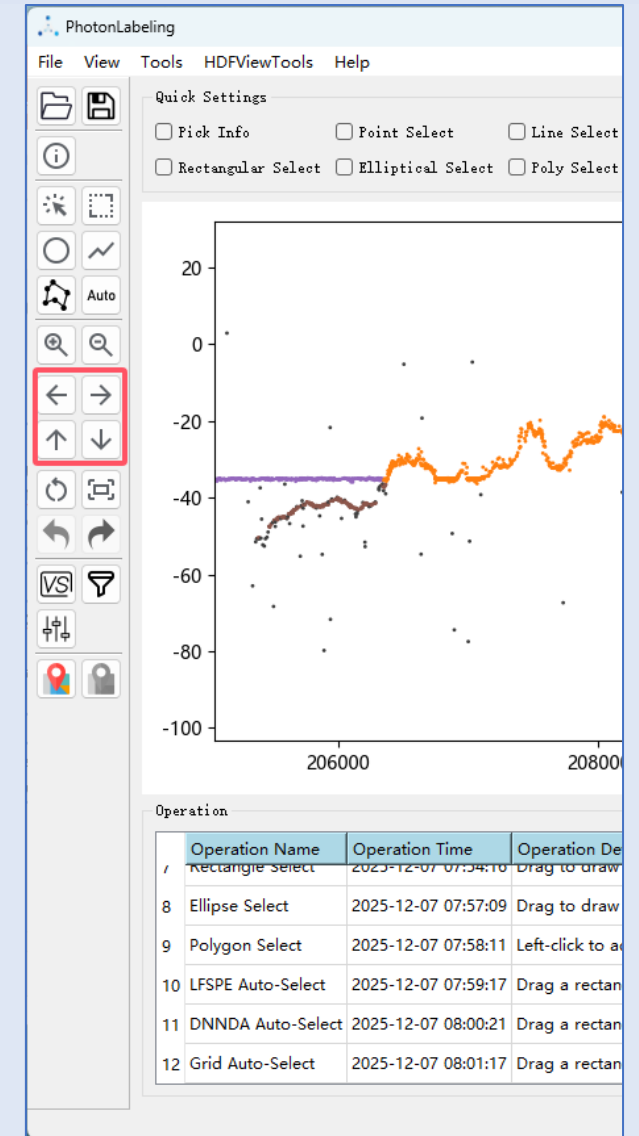
Click “Zoom Out” The main view will zoom out to show a larger area of the point cloud.

Use the mouse wheel to zoom in and out.



Click “Move Left” , “Move Right” , “Move Up” , “Move Down” The point cloud in the main window will shift accordingly.

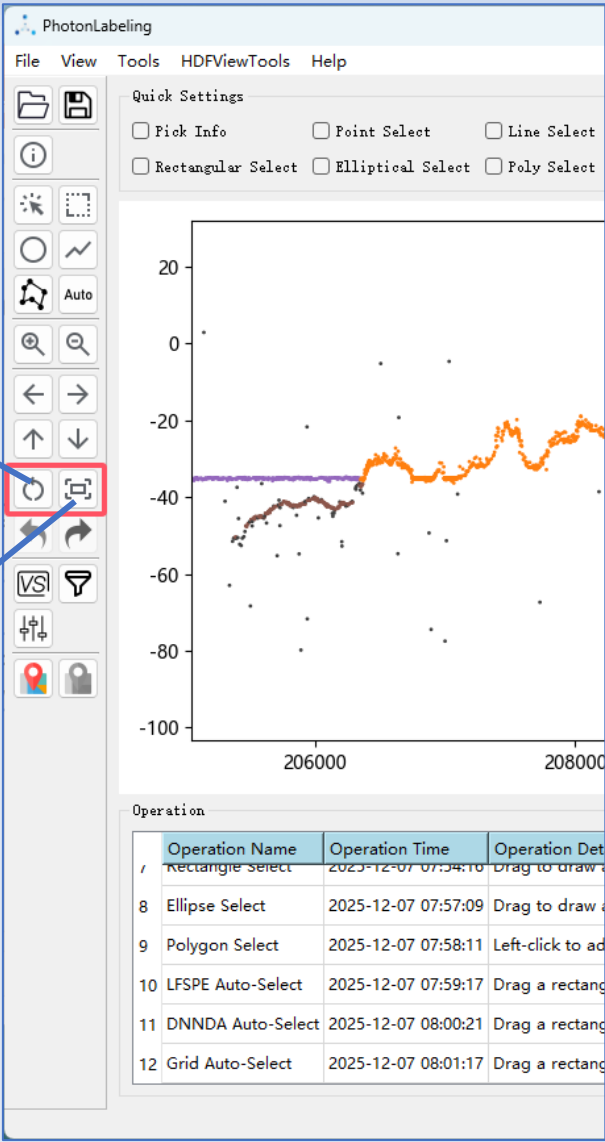
Hold down the mouse wheel (middle button) to pan the point cloud.



Tool: View Scale & Label History

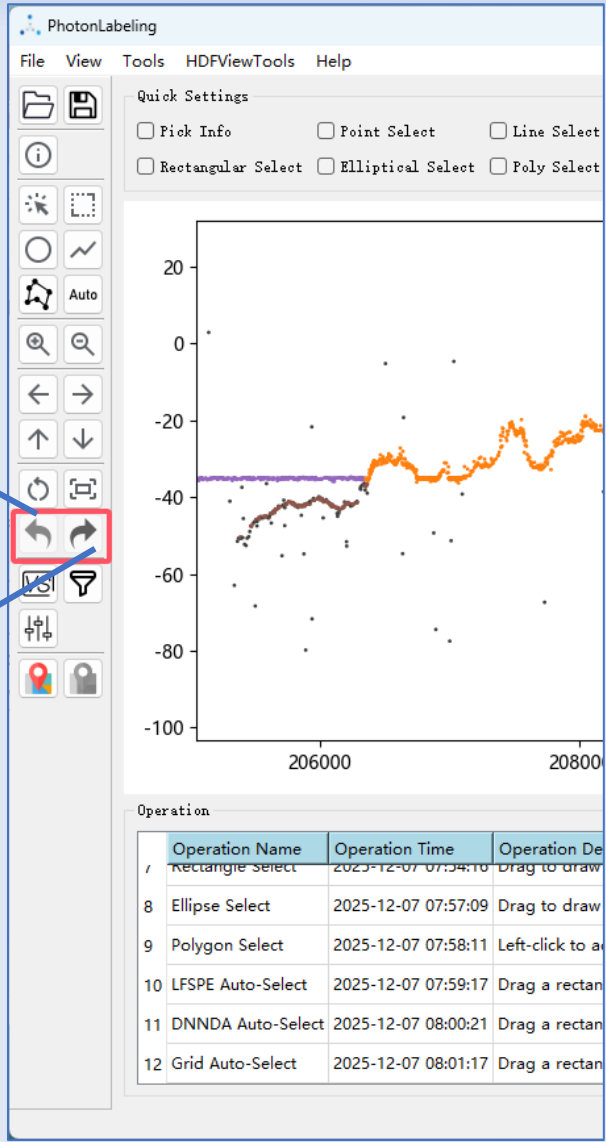
Non-uniform
Scale View

Equal Scale
View



Undo
Labeling

Redo
Labeling



Tool: VS Comparison

The “VS Comparison” tool compares the labeled results produced by PhotonLabeling with a corresponding reference dataset, and computes evaluation metrics such as:

OA – Overall Accuracy

P – Precision

R – Recall

F1 – F1-score

These metrics provide a quantitative assessment of labeling quality and the performance of different algorithms.

Quick Settings

☐ Pick Info ☐ Point Select ☐ Line Select 10.20 (Distance to Line) Point Size: 1 ☐ Rectangular Select ☐ Elliptical Select ☐ Poly Select ☐ LFSPE Auto-Select ☐ DNNDA Auto-Select ☒ Grid Auto-Select

CSV Data Comparison - Classification Metrics

Compare classification results between reference data and algorithm-labeled data. This tool will calculate Precision, Recall, F1-score, and Overall Accuracy.

File Selection

Reference Data: Select reference data CSV file...

Algorithm Data: Select algorithm-labeled CSV file...

Data Mapping Settings

Reference X column:

Reference Y column:

Reference Label column:

Algorithm X column:

Algorithm Y column:

Algorithm Label column:

Start row (0-based): 0

Number of rows: 100000

Coordinate matching tolerance: 0.001

Data Information

Number of Points: 4000
Min X: 205051.70
Max X: 214484.90
Min Y: -103.38
Max Y: 31.89

Label Settings

	Label	Name	Points
1	☒ -1	Noise	587
2	☒ 0	Signal	0
3	☒ 1	Ground Surface	910
4	☒ 2	Vegetation	0
5	☒ 3	Building	0
6	☒ 4	Water surface	1190
7	☒ 5	Underwater ...	1313
8	☒ 6	Ice	0
9	☒ 7	Other	0

Operation

Operation Name	Operation Time
Rectangle Select	2025-12-07 07:59:17
Ellipse Select	2025-12-07 07:59:17
Polygon Select	2025-12-07 07:59:17
LFSPE Auto-Select	2025-12-07 07:59:17
DNNDA Auto-Select	2025-12-07 08:00:21
Grid Auto-Select	2025-12-07 08:01:17

Tool: Label Filter

The “Label Filter” tool allows you to selectively visualize photons of specific labels. For example, you can:

Show only signal photons

Show only ground surface points

Hide certain noise or unwanted labels

The screenshot displays the PhotonLabeling software interface. The main window shows a 3D point cloud visualization of a terrain. A red arrow points from the 'Label Filter' tool icon in the left toolbar to the 'Label Filter - Select labels to display' dialog box. The dialog box is open, showing a list of labels to be displayed. The 'Label Statistics' section indicates a total of 4000 points and 4 unique labels. The list of labels includes:

- ☐ Label -1: 587 points (14.7%)
- ☐ Label 1: 910 points (22.8%)
- ☐ Label 4: 1190 points (29.8%)
- ☐ Label 5: 1313 points (32.8%)

The dialog box also features buttons for 'Select All', 'Deselect All', 'Invert Selection', 'Reset Filter', 'OK', and 'Cancel'.

Data Information

Number of Points: 4000
Min X: 205051.70
Max X: 214484.90
Min Y: -103.38
Max Y: 31.89

Label Settings

	Label	Name	Points
1	☒ -1	Noise	587
2	☒ 0	Signal	0
3	☒ 1	Ground Surface	910
4	☒ 2	Vegetation	0
5	☒ 3	Building	0
6	☒ 4	Water surface	1190
7	☒ 5	Underwater ...	1313
8	☒ 6	Ice	0
9	☒ 7	Other	0

Operation

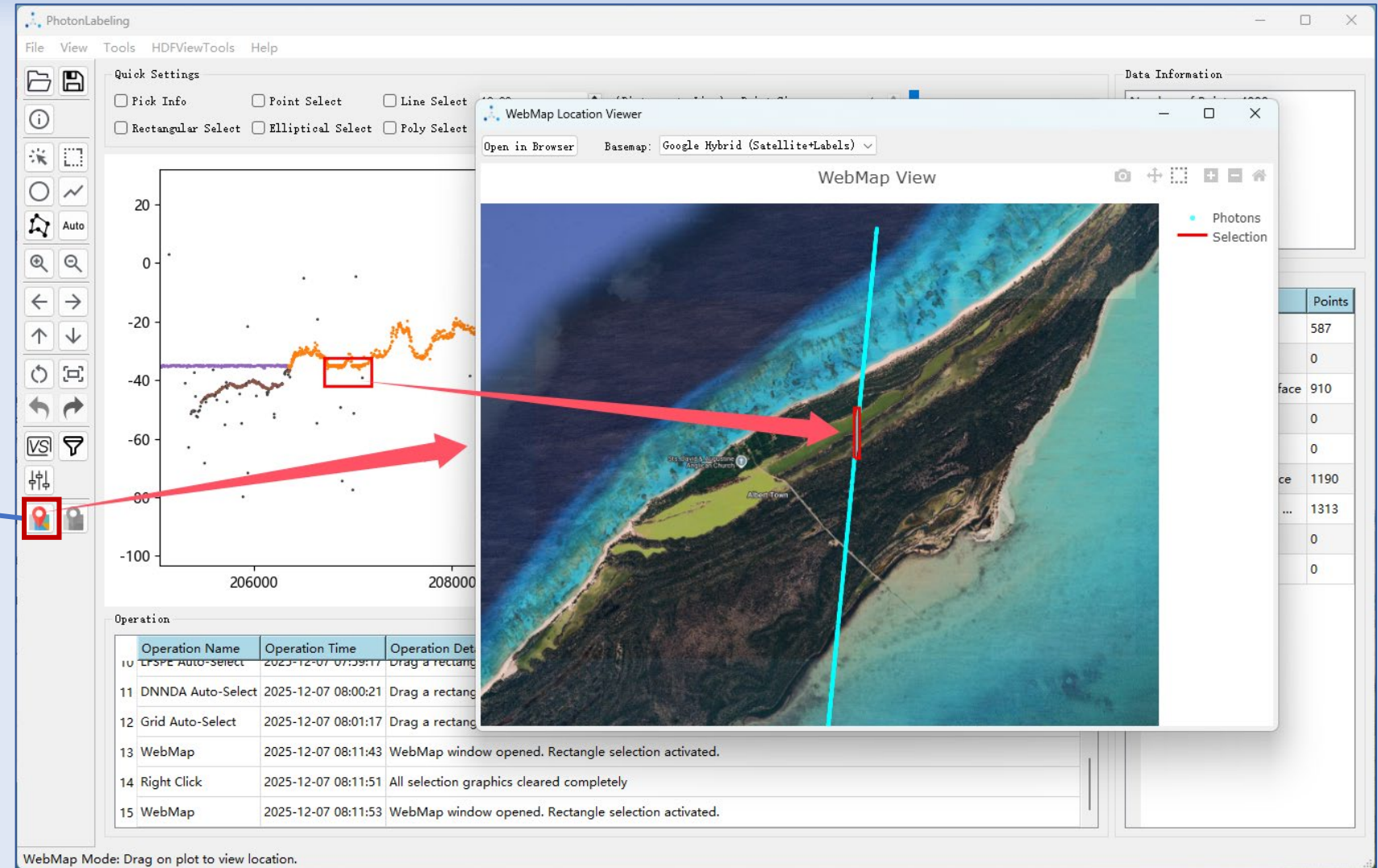
Operation Name	Operation Time	Operation Description
Rectangle Select	2025-12-07 07:34:10	Drag to select a rectangle.
Ellipse Select	2025-12-07 07:57:09	Drag to select an ellipse.
Polygon Select	2025-12-07 07:58:11	Left-click to add vertices, right-click to close polygon.
LFSPE Auto-Select	2025-12-07 07:59:17	Drag a rectangle, set parameters, then run labeling on the selection.
DNNDA Auto-Select	2025-12-07 08:00:21	Drag a rectangle to run labeling immediately on the selection; right-click to cancel.
Grid Auto-Select	2025-12-07 08:01:17	Drag a rectangle to run labeling immediately on the selection; right-click to cancel.

Tool: WebMap

The “WebMap” tool projects photon data containing longitude and latitude information onto an online web map, and displays the photon track.

Supported basemaps (for example) include common web imagery layers.

You can select regions in both the point cloud window and the map window, and the software will highlight corresponding photons in both views.



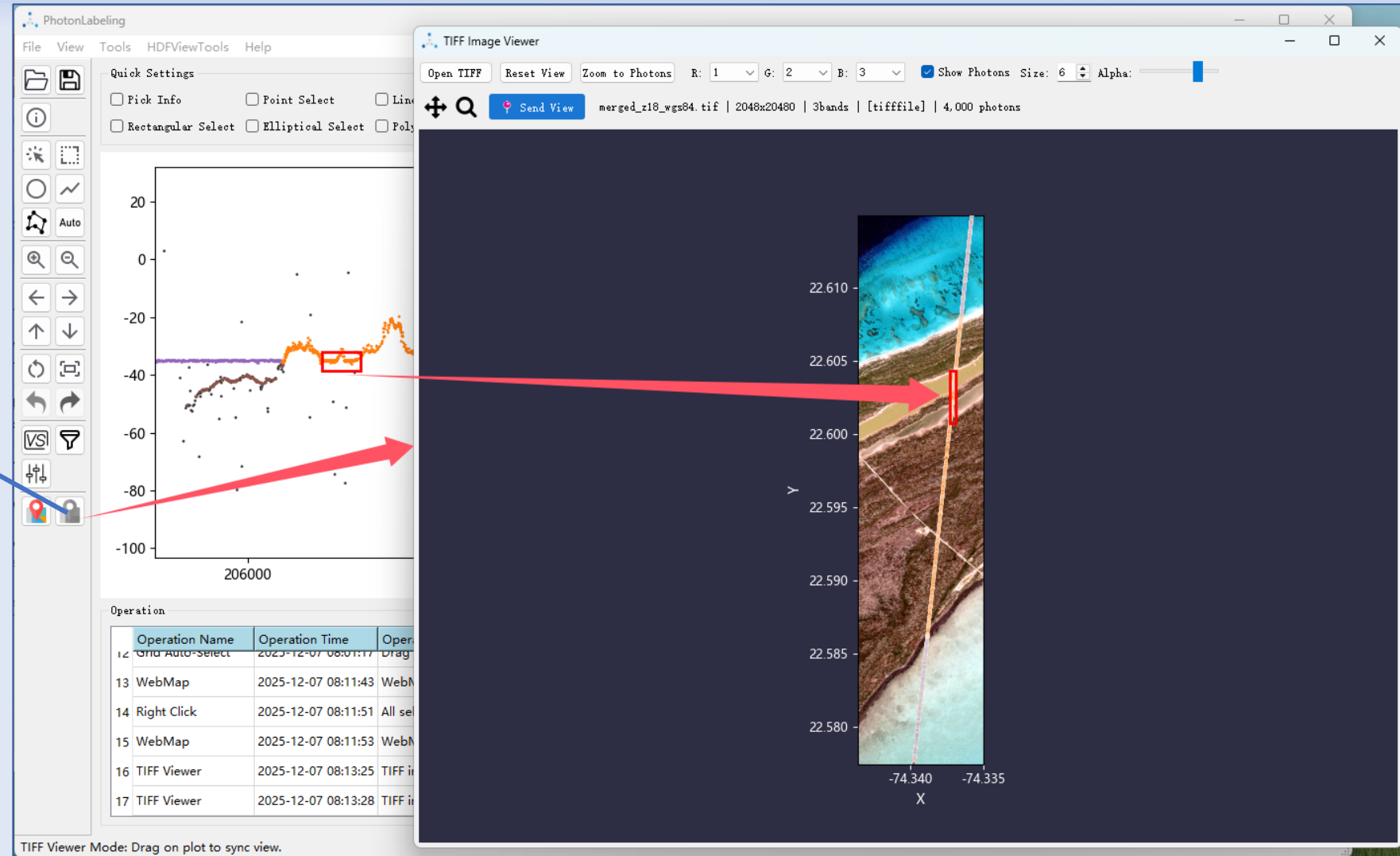
Tool: Offline Map

The “Offline Map” tool projects photon data with longitude and latitude information onto a GeoTIFF image.

Similar to WebMap, you can make selections in both the point cloud window and the map window.

The corresponding photon locations will be mutually highlighted between the two windows.

This is suitable for environments without web access or when working with custom remote-sensing imagery.



HDF Preview Tool → HDF to CSV

From the menu, choose “HDF Preview Tool → HDF to CSV” to convert HDF5 files:

Select the H5 file to be converted.

Browse and search for the content (datasets) you want to export.

Click to select the datasets to be exported.

Export the selected data to CSV format.

PhotonLabeling

FileViewToolsHelp

Quick Settings

☐ Pick Info

☐ Point Select

☐ Line Select

☐ Rectangular Select

☐ Elliptical Select

☐ Poly Select

☐ LFSPE Auto-Select

☐ DNDA Auto-Select

☐ Grid Auto-Select

10.20

(Distance to Line)

Point Size: 1

20

0

-20

-40

-60

-80

-100

206000

208000

21

0

-20

-40

-60

-80

-100

214000

Operation

Operation Name	Operation Time	Operation Details
Grid Auto-Select	2025-12-07 08:01:17	Drag a rectangle
WebMap	2025-12-07 08:11:43	WebMap window
Right Click	2025-12-07 08:11:51	All selection graphics cleared completely
WebMap	2025-12-07 08:11:53	WebMap window opened. Rectangle selection activated.
TIFF Viewer	2025-12-07 08:13:25	TIFF image viewer opened with 4000 photon points.
TIFF Viewer	2025-12-07 08:13:28	TIFF image viewer opened with 4000 photon points.

Select Datasets to Export

Load HDF5 File

heights

gt1l/heights/delta_time

gt1l/heights/dist_ph_across

gt1l/heights/dist_ph_along

gt1l/heights/h_ph

gt1l/heights/lat_ph

gt1l/heights/lon_ph

gt1l/heights/pce_mframe_cnt

gt1l/heights/ph_id_channel

gt1l/heights/ph_id_count

gt1l/heights/ph_id_pulse

gt1l/heights/quality_ph

gt1l/heights/signal_conf_ph

gt1r/heights/delta_time

gt1r/heights/dist_ph_across

gt1r/heights/dist_ph_along

gt1r/heights/h_ph

gt1r/heights/lat_ph

gt1r/heights/lon_ph

gt1r/heights/pce_mframe_cnt

gt1r/heights/ph_id_channel

gt1r/heights/ph_id_count

gt1r/heights/ph_id_pulse

Export to CSV

Data Information

Number of Points: 4000

Min X: 205051.70

Max X: 214484.90

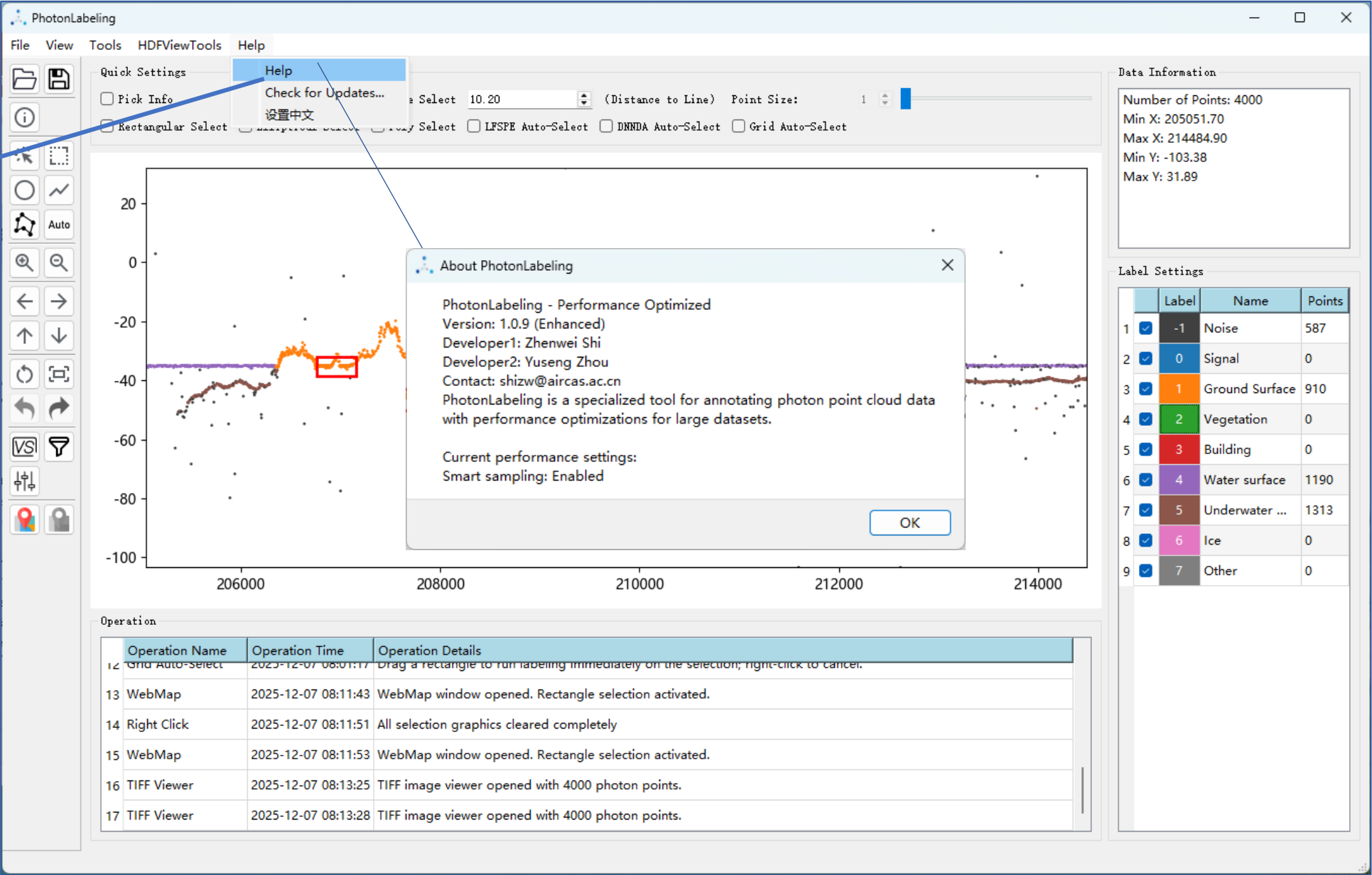
Min Y: -103.38

Max Y: 31.89

Label Settings

	Label	Name	Points	
1	☒	-1	Noise	587
2	☒	0	Signal	0
3	☒	1	Ground Surface	910
4	☒	2	Vegetation	0
5	☒	3	Building	0
6	☒	4	Water surface	1190
7	☒	5	Underwater ...	1313
8	☒	6	Ice	0
9	☒	7	Other	0

Help Menu



- ❑ View Help – Open the user guide.
- ❑ Check for Updates – Check whether a newer version of the software is available.
- ❑ Switch Language (Chinese / English) – Toggle the software interface between Chinese and English.

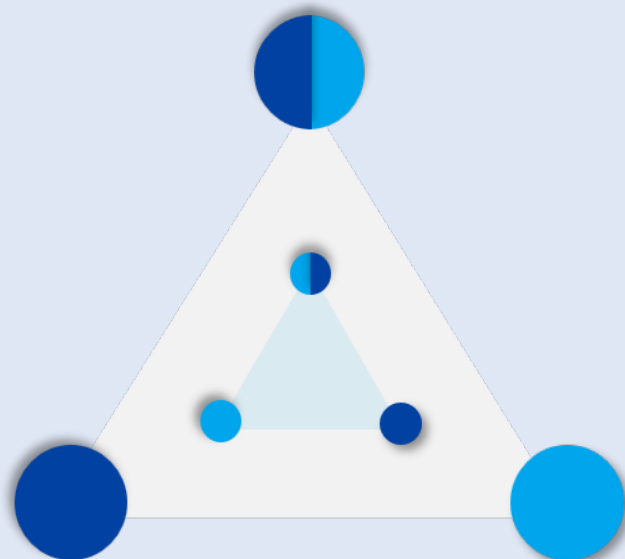


中国科学院空天信息创新研究院

Aerospace Information Research Institute, Chinese Academy of Sciences

Photon Labeling

Software User Manual



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Modify or plagiarize the software

Tamper with copyright information

Perform reverse engineering or similar operations

Feedback and Support

This software may still contain limitations or missing features. Users from all fields are welcome to provide comments, criticism, and suggestions so that we can continually improve and optimize the software.

Feedback / Issues: shizw@aircas.ac.cn

Download: <https://github.com/zwshi-pku/PhotonLabeling>